

4.3 Contrastive Learning and Zero-Shot Transfer

InfoNCE wants the diagonal. Zero-shot is nearest prompt, not a trained C -way head.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Contrastive learning pulls matched pairs together and pushes unmatched pairs apart, in a shared space. **Zero-shot** transfer is what you get when one side of the pair can be a class name written as text. This is a **coordination** loss: two towers, a similarity, no fused decoder.

Week 5 will put CLIP's data and retrieval on the table. This note is the loss and the zero-shot trick.

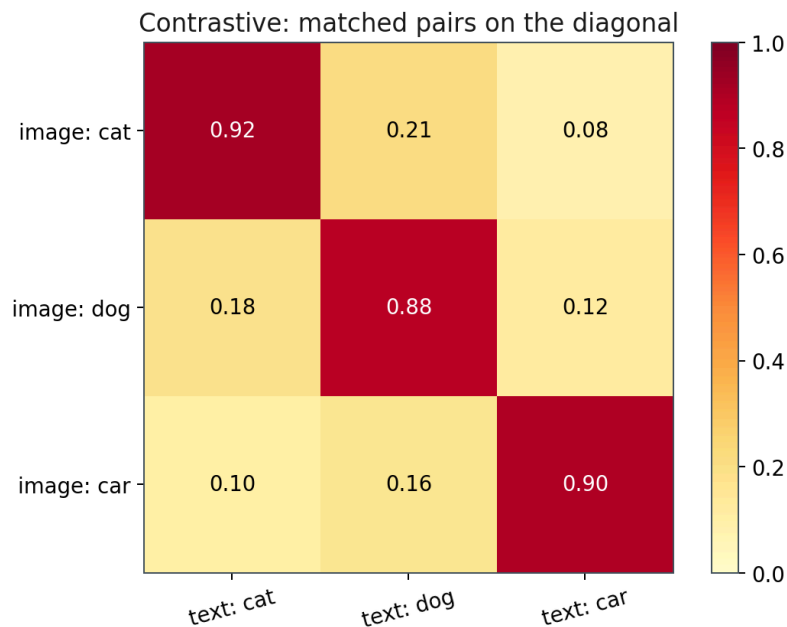
First time this method appears. **Contrastive learning** pulls matched pairs together and pushes the rest apart. **Zero-shot** is classify by nearest prompt, not a C -way trained head.

What. InfoNCE on a batch of N matched pairs. Zero-shot: embed class phrases, pick $\arg \max_c \cos(f(x), g(\text{prompt}_c))$. **Why.** You may not have labels for every class at train time. A similarity space transfers to new names. **Architecture.** Two encoders (or two views of one). A batch matrix of scores. Softmax over the row (or bidirectional). **How.** Lab 4: four pairs, heatmap of the $N \times N$ scores. The diagonal should win. **Formula.** $\mathcal{L} = -\frac{1}{N} \sum_i \log \frac{\exp(s_{ii} / \tau)}{\sum_j \exp(s_{ij} / \tau)}$, $s_{ij} = \cos(u_i, v_j)$.

Tradeoffs. + No C -way head, works with prompts. – Needs big batches (many negatives); temperature τ is a knob; compositionality is not free.

1. A batch of matches

N images and N captions. Encode both. Cosine similarities form an $N \times N$ matrix. The loss (InfoNCE; Oord et al. 2018) wants the **diagonal** hot: image i with caption i , not caption j .



This is coordinated representation (note **4.1**) with a specific loss. You never fuse pixels and words into one vector during pretraining; you only score pairs. SimCLR is the vision-only cousin (two crops of the same image). We cite it; we do not assign it.

InfoNCE, one direction, row i :

$$-\log \frac{\exp(\text{sim}(v_i, t_i) / \tau)}{\sum_{j=1}^N \exp(\text{sim}(v_i, t_j) / \tau)}.$$

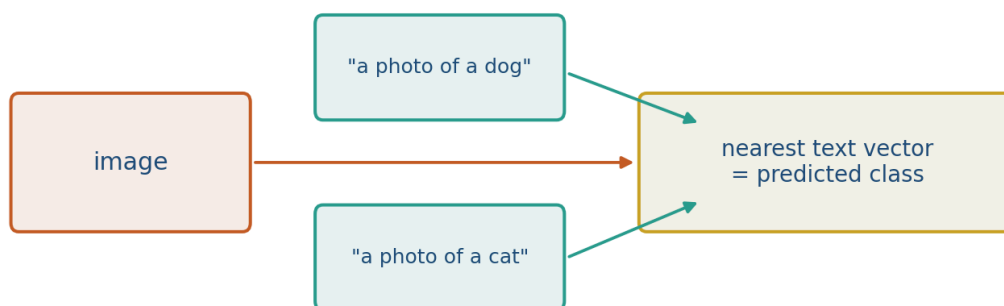
CLIP uses **both** directions (image-to-text and text-to-image). Temperature τ sharpens the softmax. Small τ makes the diagonal even more peaked, but the **argmax** of a row does not change if you only rescale that row.

The negative set is the rest of the batch. Small batches $\Rightarrow$ easy negatives $\Rightarrow$ a weak space. CLIP's real batch was tens of thousands. Lab 4 is $N = 4$ so you can see the matrix, not so you can match ImageNet.

2. Zero-shot classification

At test time, encode the image once. Encode prompts "a photo of a dog", "a photo of a cat", ... Take the nearest prompt. There is no trained softmax over ImageNet. New classes are new strings.

Zero-shot: class names are text embeddings, not trained heads



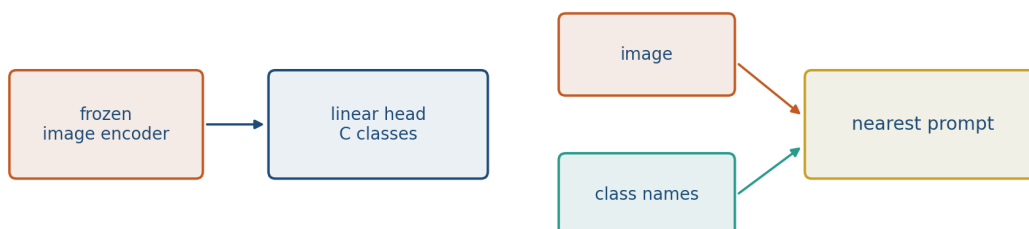
CLIP was trained on **phrases**. The raw label "dog" often loses to "a photo of a dog". Prompt **ensembling** (average several phrasings) is a test-time trick, not a new model. We mention it; Lab 4 does not require it.

A class the text tower never saw as a phrase will not magically appear. Open-vocabulary is not infinite-vocabulary.

3. Linear probe versus zero-shot

Linear probe: train a softmax

Zero-shot: no trained C-way head



Linear probe: freeze the image encoder, train a C -way softmax on labeled images. Needs a training set for those C classes.

Zero-shot: no extra weights. The classifier is the text tower. Adding a class is adding a string.

Both are transfer. Zero-shot is what makes CLIP a **foundation** model. Captioning, LLaVA, and Flamingo wait for later weeks.

Web pairs are noisy (Week 5: BLIP filters them). Contrastive spaces can still be **biased** (Week 5.3).

4. Teaching this note

~18 minutes. Draw the $N \times N$ grid, mark the diagonal, write InfoNCE for $N = 2$, then swap the text side for class prompts, then probe vs zero-shot. Play the CLIP video on **zero-shot** and **the similarity matrix**

(9:00–22:25). Lab 4’s tiny InfoNCE is this grid with $N = 4$.

5. Worked example

Two pairs. Cosine matrix (already normalized):

$$S = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}, \quad \tau = 1.$$

Row 0 softmax: $e^{0.9} \approx 2.46$, $e^{0.1} \approx 1.11$, so $[0.69, 0.31]$. InfoNCE term $-\log 0.69 \approx 0.37$.

InfoNCE: the rest of the batch are negatives. The diagonal must be special.

row 0 scores	softmax	$-\log p_{\text{diag}}$
$[0.9, 0.1], \tau = 1$	$[0.69, 0.31]$	0.37
identical captions	uniform	cannot learn

If every caption is the same vector, every row of S is constant, softmax is uniform, and the diagonal is not special: **the loss cannot learn matching**.

Zero-shot: image $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, prompts $\text{dog} = \begin{bmatrix} 0.9 \\ 0.1 \end{bmatrix}$, $\text{cat} = \begin{bmatrix} 0.1 \\ 0.9 \end{bmatrix}$. Cosines 0.99 vs 0.10. Predict dog.

Dividing by $\tau = 0.07$ before softmax does **not** change that argmax.

6. Where students get stuck

- Treating off-diagonal as “don’t care” instead of **negatives**.
 - Adding a class at test time that the text tower cannot spell.
 - Using batch size 4 in a real CLIP run and expecting ImageNet-level negatives.
 - Calling zero-shot a trained ImageNet head.
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7. Video

Watch [Yannic Kilcher: OpenAI CLIP, Connecting Text and Images](#).

Pause on **zero-shot (9:00)** — class names as text — and on **InfoNCE / the $N \times N$ matrix (14:40)**. That matrix is the whole note.

8. Practice

1. If every caption in the batch is the word *photo*, what happens to the diagonal?
2. Why is "a photo of a golden retriever" often better than "golden retriever" for CLIP-style zero-shot?
3. Name one class you **cannot** add at test time by typing it.
4. For $S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\tau = 1$, write the two InfoNCE row-losses (they should be 0). Now change S_{01} to 1. Recompute row 0's softmax and $-\log p_{\text{diag}}$.
5. Three prompts with cosines to one image 0.2, 0.5, 0.1. Which class wins zero-shot? If you divide the cosines by $\tau = 0.07$ before softmax, does the **argmax** change?
6. Linear probe versus zero-shot: which one needs a labeled training set for the C classes?
7. CLIP uses both image-to-text and text-to-image InfoNCE. Why is one direction not enough if you also want to retrieve images from a caption?